

Test case

- This document contains detailed test cases for the JobLaunch web-based job application system. Each case maps to the functional and non-functional requirements from the SRS and the design artifacts (API endpoints, database schema, and UI mockups). The goal is to validate core features and quality attributes before the Phase 4 submission.

Test case format

Each test case includes the following fields:

➔ Test Case ID	➔ Preconditions	➔ Expected Result
➔ Requirement ID	➔ Test Data	➔ Priority
➔ Title	➔ Steps	➔ Status

A. Authentication and authorization

TC-AUTH-01

Requirement ID: FR-AUTH-01

Title: Register as job seeker

Preconditions: No existing account with the test email

Test Data: full_name="Test Seeker", email="seeker@test.com", password="StrongPass!23", role="Seeker"

Steps:

1. Open signup page
2. Fill form with test data
3. Submit form

Expected Result: Account is created, confirmation shown, login available

Priority: High

Status: Pending

TC-AUTH-02

Requirement ID: FR-AUTH-01

Title: Register as employer

Preconditions: No existing account with the test email

Test case

Test Data: full_name="Test Employer", email="employer@test.com", password="StrongPass!23", role="Employer"

Steps:

1. Open signup page
2. Fill form with employer data
3. Submit form

Expected Result: Employer account is created, confirmation shown, login available

Priority: High

Status: Pending

TC-AUTH-03

Requirement ID: FR-AUTH-03

Title: Login issues JWT on success

Preconditions: Account exists for seeker@test.com

Test Data: email="seeker@test.com", password="StrongPass!23"

Steps:

1. Open login page
2. Enter valid credentials
3. Submit

Expected Result: Login succeeds; JWT issued and stored (secure cookie or secure storage); redirect to dashboard

Priority: High

Status: Pending

TC-AUTH-04

Requirement ID: FR-AUTH-02

Title: Password stored hashed

Preconditions: New account created

Test Data: Inspect DB via admin console or API

Steps:

1. Create user
2. Query Users table for password field

Test case

Expected Result: password_hash is non-reversible hash (no plaintext)

Priority: High

Status: Pending

TC-AUTH-05

Requirement ID: NFR-07

Title: Role-based access control prevents cross-tenant access

Preconditions: Two employers exist with different jobs and applicants

Test Data: Employer A JWT, Employer B job IDs

Steps:

1. Call "GET applicants" for Employer B's job using Employer A's JWT

Expected Result: Access denied with appropriate error; no data leaked

Priority: High

Status: Pending

B. Job management (employer)

TC-JOB-01

Requirement ID: FR-JOB-01

Title: Create job posting with required fields

Preconditions: Logged in as Employer

Test Data: title, description, salary range, location, category

Steps:

1. Open Create Job form
2. Fill all required fields
3. Submit

Expected Result: Job record created; visible in employer's job list and public search

Priority: High

Status: Pending

TC-JOB-02

Requirement ID: FR-JOB-02

Test case

Title: Edit own job posting

Preconditions: Existing job created by logged-in employer

Test Data: Updated title or description

Steps:

1. Open Edit Job page for owned job
2. Modify fields
3. Save changes

Expected Result: Job updated; changes reflected in details and search results

Priority: High

Status: Pending

TC-JOB-03

Requirement ID: FR-JOB-02

Title: Delete own job posting

Preconditions: Existing job created by logged-in employer

Test Data: Job ID

Steps:

1. Open Job management page
2. Delete job

Expected Result: Job removed; not shown in lists and search; applications remain or handled per design

Priority: Medium

Status: Pending

TC-JOB-04

Requirement ID: NFR-07

Title: Prevent editing jobs owned by other employers

Preconditions: Job belongs to Employer B; logged in as Employer A

Test Data: Employer A JWT, Employer B job ID

Steps:

1. Attempt PUT /api/jobs/:id with Employer A JWT for Employer B job

Test case

Expected Result: Authorization error; no update applied

Priority: High

Status: Pending

C. Search and browse (seeker)

TC-SEARCH-01

Requirement ID: FR-SEARCH-01, NFR-06

Title: Keyword search returns results within 2 second

Preconditions: Multiple jobs exist with known keywords

Test Data: query="Frontend Engineer"

Steps:

1. Enter keyword in search field
2. Submit or typeahead search

Expected Result: Matching jobs displayed; response time under 2 second

Priority: High

Status: Pending

TC-SEARCH-02

Requirement ID: FR-SEARCH-01

Title: Filter by company name and category

Preconditions: Jobs from multiple companies and categories exist

Test Data: company="Acme Ltd", category="Remote"

Steps:

1. Apply company and category filters

Expected Result: Only jobs from selected company and category are shown

Priority: Medium

Status: Pending

TC-SEARCH-03

Requirement ID: FR-SEARCH-02

Test case

Title: View job details page

Preconditions: A job exists

Test Data: Job ID

Steps:

1. Click a job card

Expected Result: Detailed view shows title, description, salary, location, category, and posted date

Priority: Medium

Status: Pass

D. Application workflow

TC-APP-01

Requirement ID: FR-APP-01

Title: Submit application with optional message

Preconditions: Logged in as Seeker; job exists

Test Data: resume_url or file, cover_letter text

Steps:

1. Open job details
2. Click Apply
3. Upload resume, add optional message
4. Submit

Expected Result: Application created; success confirmation shown

Priority: High

Status: Pending

TC-APP-02

Requirement ID: FR-APP-02, DB constraint UNIQUE(job_id, seeker_id)

Title: Prevent duplicate applications to same job

Preconditions: One application already exists for job by same seeker

Test Data: Same job_id, same seeker

Steps:

Test case

1. Attempt to apply again

Expected Result: Request rejected; error message; no duplicate record created

Priority: High

Status: Pending

TC-APP-03

Requirement ID: FR-APP-03

Title: Employer updates application status

Preconditions: Employer logged in; applications exist for their job

Test Data: status transitions = Pending → Interview Scheduled → Rejected

Steps:

1. Open applicants list
2. Select an application
3. Update status

Expected Result: Status updated; visible in employer and seeker views

Priority: High

Status: Pending

TC-APP-04

Requirement ID: FR-APP-04

Title: Seeker views application status history

Preconditions: Seeker has submitted one or more applications with status updates

Test Data: Seeker JWT

Steps:

1. Open “My Applications” page

Expected Result: List shows job, date, current status; details available

Priority: Medium

Status: Pending

E. Security and session management

Test case

TC-SEC-01

Requirement ID: Session security design

Title: JWT stored securely

Preconditions: Successful login

Test Data: Inspect storage mechanism

Steps:

1. Log in
2. Verify JWT storage (secure cookie or secure storage)

Expected Result: JWT is not accessible via JavaScript if using httpOnly cookies; otherwise protected with best practices

Priority: High

Status: Pending

TC-SEC-02

Requirement ID: NFR-07

Title: Unauthorized access blocked on protected endpoints

Preconditions: No JWT or invalid JWT

Test Data: Call protected API without token

Steps:

1. Request protected API

Expected Result: 401 Unauthorized or equivalent; no sensitive data returned

Priority: High

Status: Pending

F. Performance

TC-PERF-01

Requirement ID: NFR-05

Title: CRUD API executes under 1.5 seconds

Preconditions: Staging environment running; DB populated

Test Data: Create, Read, Update, Delete on Jobs and Applications

Test case

Steps:

1. Measure API latency for each CRUD operation using Postman or logs

Expected Result: Each operation completes in under 1.5 seconds

Priority: High

Status: Pending

TC-PERF-02

Requirement ID: NFR-06

Title: Search results under 2 second

Preconditions: Jobs indexed; normal load

Test Data: Common keyword search

Steps:

1. Trigger search

Expected Result: Results displayed in under 1 second

Priority: High

Status: Pending

G. Usability and responsiveness

TC-UX-01

Requirement ID: NFR-09

Title: Responsive layout across desktop and mobile

Preconditions: Deployed frontend with CSS

Test Data: Chrome DevTools device modes

Steps:

1. Open key pages (Home, Login, Job Details, Apply)

2. Test at multiple breakpoints

Expected Result: Layouts adapt cleanly; no overlap or hidden controls

Priority: Medium

Status: Pass

TC-UX-02

Requirement ID: NFR-10

Title: Client-side validation and consistent server messages

Test case

Preconditions: Forms implemented

Test Data: Invalid inputs (empty fields, invalid email, short password)

Steps:

1. Submit forms with invalid data

Expected Result: Clear client-side validation messages; server errors match tone and content when triggered

Priority: Medium

Status: Pass

H. Data integrity and database constraints

TC-DB-01

Requirement ID: Applications unique constraint

Title: Enforce UNIQUE(job_id, seeker_id)

Preconditions: One application exists for job

Test Data: Attempt second insert for same job and seeker

Steps:

1. Submit second application

Expected Result: DB or API rejects with conflict; no duplicate stored

Priority: High

Status: Pending

TC-DB-02

Requirement ID: Foreign keys and referential integrity

Title: Prevent orphaned applications

Preconditions: Application exists for a job

Test Data: Delete job

Steps:

- i. Attempt to delete job with existing applications

Expected Result: Either cascade behavior per design or prevention with clear error; no orphaned rows

Priority: Medium

Test case

Status: Pending

ii. Test data management

Use distinct emails and UUIDs per scenario. Clean up created jobs and applications after tests. Maintain a small seed dataset for search tests.

iii. Bug logging and reporting

Log all defects in GitHub Issues with the following fields: Title, Steps to Reproduce, Expected vs Actual, Severity, Assignee, Status. Perform regression testing after fixes.

iiii. Exit criteria

All high-priority test cases pass, critical and major defects resolved, prototype stable for demo.